Research Project Instructions

Goal: Based on your individual mini literature review, you will now pair up (2–3 students) and design a joint research project. The purpose is to combine your findings and fill one of the identified gaps by formulating and addressing a concrete research question. The project may be theoretical or empirical in nature.

Scope and format

- **Max length:** 6 pages (references do not count toward the limit), 11 pt, single-spaced.
- **Collaboration:** Groups of 2–3 students.
- **Report structure:**

1. **Introduction:** motivate the area, summarize combined findings from the reviews.
2. **Research Problem and Question:** Define the gap, clearly state the research question / sub-questions for the niche you are going to focus on. Recall that we don't expect you to invent a new algorithm or method in a month.

   Your project, for instance, could also focus on establishing and confirming the gap by designing a study and running state-of-the-art methods, followed by analysis of the results to establish the gap or refute your hypothesis.

   Your project could also be a reproducibility study but it should lead to some interesting insight. For instance, you might want to check if factuality alignment methods like FactAlign, FLAME in general domain question answering / response generation reproduce when evaluated in medical question answering ?

   Another example could be finding a gap in evaluation metric and proposing a new metric. While we understand you may not be able to invent a successful metric in a month, if you can execute experiments showing gap in existing metrics and preliminary theory and experiments for an idea of a new metric showing potential, it is good. Make sure your study is small but focused enough to complete within a month and gives some interesting insight for you to build upon.
3. **Method:** describe how you approach the question (theoretical framework, experiment, data, evaluation). Make sure your approach is clearly described. For instance, if the core focus of your project is a study to show that retrieval is the gap in fact-checking / question-answering more than verification, you have to design a study that can demonstrate the upper bound with golden evidence and how far behind current retrieval approaches are. Apart from quantitative evaluation, you also have to analyze the responses and discover if cases where evidence retrieval has failed whether it directly impact downstream answer generation / verification. Make sure to include statistical tests wherever applicable. Recall the methodology lectures.
4. **Results:** present findings, reasoning, or initial experiments. In results, please note that you have to synthesize and explain the insights and not just present the Tables and plots. Please make sure your tables and plots are presentable and coherent. Do not spam with too many irrelevant experiments. Please note that AI frequently does this, as it is also quite verbose in planning and performs experiments that are commonly well-discussed in the literature at that point. Hence it recommends a lot of obvious or irrelevant experiments (irrelevant in sense it does not answer the core question of the study). So as re-iterated multiple times please use AI sparingly and do not let it be the driver or planner of experiments. Experiment design to investigate the research question is a core skill you have to develop.
5. **Conclusion:** reflect on the contribution, limitations, and future work.

Template You can use a double-column or single-column template from one of the following. The page limit is 6 is either case, so please choose wisely. Please choose only one of the following templates

- **ACL**: `https://www.overleaf.com/read/rrdjzjgwgbrn#6e5035`
- **ACM DL**: `https://www.overleaf.com/read/rttscgvkbwjt#7005d0`
- **IEEE**: `https://www.overleaf.com/read/fnttzpynpmcm#f1e8d6`
- **ICLR**: `https://www.overleaf.com/read/nzgfzsjtrqpk#9d9b06`

Process checklist

1. Meet with your partner(s) and compare your literature reviews.
2. Identify overlaps and complementary gaps.
3. Select one gap as the focus of your joint project.
4. Formulate a precise research problem and question.
5. Choose an appropriate method (conceptual analysis, small experiment, pilot study, etc.).
6. Collect and analyze results (theoretical argument or empirical evidence).
7. Write the 6-page report with clear division of roles and joint synthesis. Please clearly state who contributed to which aspect in the project.

Please also utilize the office hours offered by instructors

Grading rubric

Criterion	0: Insufficient	1: Adequate	2: Excellent
Integration of Reviews	No attempt to combine findings; unclear starting point.	Some synthesis, but weak integration of individual reviews.	Clear, coherent integration of reviews into a shared foundation.
Research Problem	Problem/question vague, trivial, or missing.	Problem/question stated but lacks precision or depth.	Well-defined, precise, and meaningful research problem and question.
Method	No method or inappropriate approach.	Method partially fits but lacks detail or rigor.	Appropriate, well-justified method; transparent and feasible.
Results	No results or irrelevant to the question.	Limited or unclear results; weak connection to question.	Relevant, clear results or reasoning; strongly tied to the research question.
Results Discussion	No discussion or in depth analysis of results and insights	Limited or unclear discussion with not so clear highlight of the core insights	Relevant, clear reasoning for the results observed with qualitative examples when applicable; Core insights well described synthesized from observed results ; strongly tied to the research question.
Clarity & Structure	Poorly written, unclear, disorganized, referencing inconsistent.	Mostly clear, some structural or referencing issues.	Clear, professional, logically structured, consistent references.